



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2016
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

16 September 2016

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer **one** question on the answer paper provided.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
 - Section A (Part I)
 - Section A (Part II)
 - Section B

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
Section A	
1	/ 10
2	/ 9
3	/ 10
4	/ 11
5	/ 11
6	/ 9
7	/ 11
8	/ 9
Section B	
9 / 10	/ 20
Total	/ 100

This paper consists of __ printed pages.

[Turn over]

Section A (Part I)
Answer **all** the questions in this section.

QUESTION 1

The structure of the tubulin dimer, the protein that forms microtubules by polymerisation, is shown in **Fig. 1.1**.

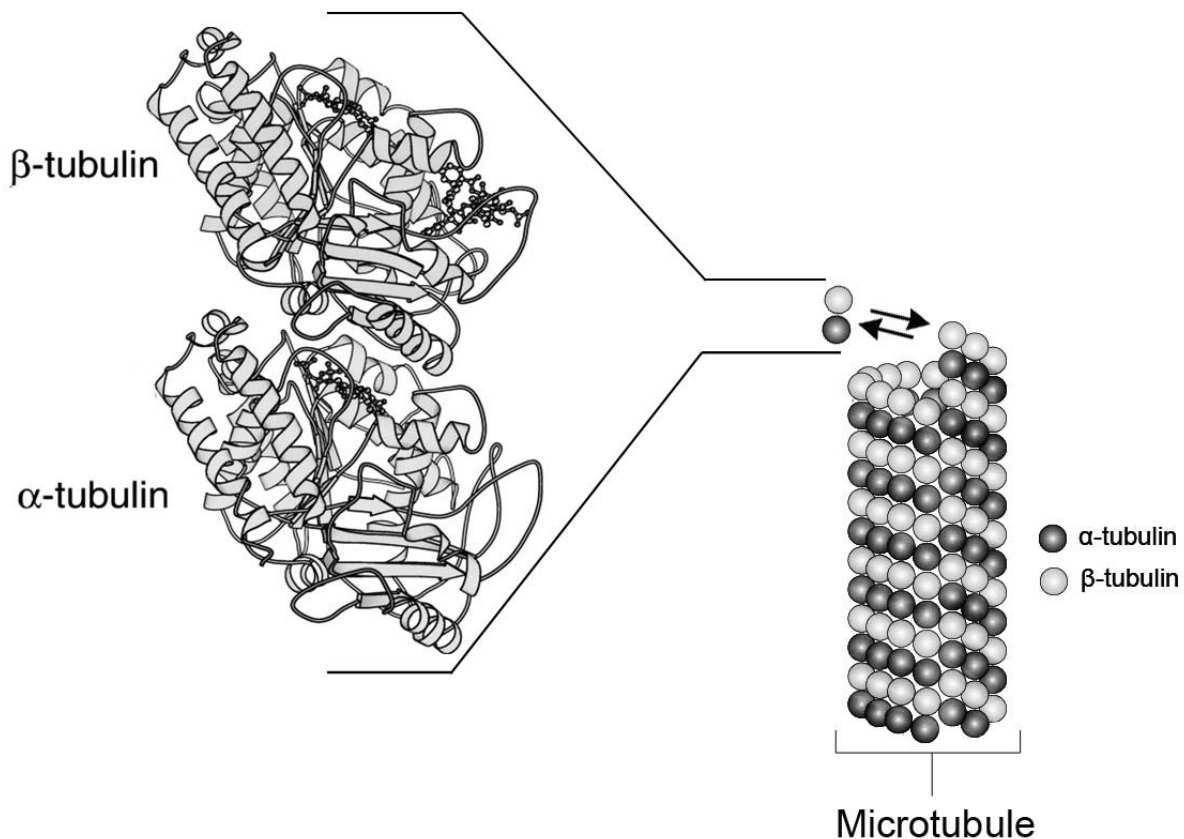


Fig. 1.1

(a) With reference to **Fig. 1.1**, name the secondary structures present in tubulin. [1]

- α -helices and β -pleated sheets.

Tubulin inhibitors like paclitaxel and vinblastine have been utilised in chemotherapy drug trials to treat cancers. All tubulin inhibitors are known to bind to the β -tubulin subunit.

(b) Explain how tubulin inhibitors reduce tumour formation. [3]

- Tubulin inhibitors interfere / prevent polymerisation of tubulin dimers to form microtubules / spindle fibres that make up the mitotic spindle.
- Without spindle fibres, chromosomes cannot divide / mitosis stops at prophase / mitosis cannot take place.
- Hence affected cells exit the cell cycle / go into G_0 and uncontrolled cell division is prevented.

A trial was conducted to compare the effects of vinblastine and paclitaxel on the SKOV3 ovarian cancer cell line and the PC3 prostate cancer cell line.

Table 1.1 below shows the results of the trial. The researcher measured the number of months in which the mass of tumours increased to critical mass after treatment with vinblastine and paclitaxel.

	No. of months in which the mass of tumours increased to critical mass	
	SKOV3	PC3
Untreated	0.5	1.0
Vinblastine	5.7	10.0
Paclitaxel	10.1	9.1

Table 1.1

(c) With reference to **Table 1.1**, compare the effects of vinblastine and paclitaxel on tumour growth in the two cancer cell lines. [4]

Similarities (max 2):

- **Both** vinblastine and paclitaxel **reduces / slowed down tumour growth** in **both** the SKOV3 and PC3 cells lines. [Coupled with data citation]
- Vinblastine and paclitaxel were **equally effective** in slowing down tumour growth in **PC3** tumours.
 - o Data citation: **Untreated SKOV3 tumours** grew critical mass in **0.5 months** as compared to tumour cells treated with **vinblastine** which only grew to critical mass after **5.7 months** and tumour cells treated with **paclitaxel** which only grew to critical mass after **10.1 months**.
 - o Data citation: **Untreated PC3 tumours** grew to critical mass in **1.0 months** as compared to tumour cells treated with **vinblastine** which only grew to critical mass after **10.0 months** and tumour cells **treated** with **paclitaxel** which only grew to critical mass after **9.0 months**.

Differences (max 2):

- Vinblastine **slowed down tumour growth to a lesser extent / was less effective in treating the SKOV3 cell lines** as compared to paclitaxel. [Coupled with data citation]
 - o SKOV3 tumour cells **treated** with **vinblastine** grew to critical mass in **5.7 months** as compared to SKOV3 cells treated with **paclitaxel** which only grew to critical mass after **10.1 months**.

(d) Suggest and explain why different tumour cells may exhibit different levels of resistance to the same drug. [2]

- Different tumour cells exhibit **variation due to random mutation** to certain genes / **differential gene expression / gene amplification**.
- Leading to formation of new alleles coding for / more functional proteins that serve as
 - o **transporter proteins** that **pump out** drugs.
 - o **Tubulin** with a non-complementary **binding site** where the drug usually binds to.
 - o **enzymes** that **hydrolyses** the drug.
 - o **inhibitors** that bind to the drugs and render them ineffective.
 - o AVP, e.g. cell surface receptors, proteins that regulate the cell cycle, etc

[Total: 10]

QUESTION 2

Fig. 2.1 shows the process of translation.

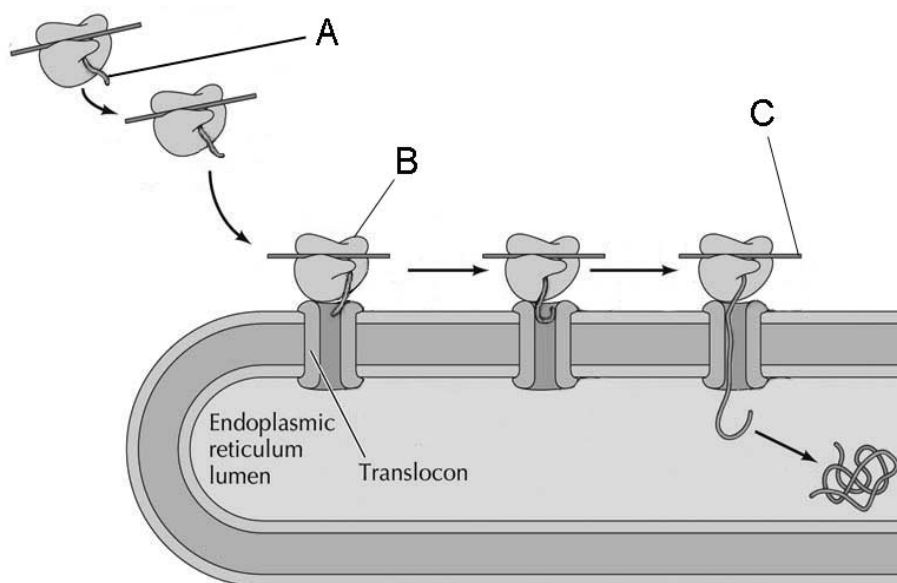


Fig. 2.1

(a) (i) Label structures **A**, **B** and **C**.

[3]

- A – polypeptide chain
- B – small ribosomal subunit (accept ribosome)
- C – mRNA

(ii) Suggest the role of the translocon in protein synthesis.

[1]

- The translocon serves as a **hydrophilic channel** to allow the passage of the polypeptide chain into the **lumen** of the endoplasmic reticulum.

(b) List two ways in which transcription differs from DNA replication.

[2]

	DNA Replication	Transcription
Template	Both strands of DNA	Template strand / one of two DNA strands
Raw materials	Deoxyribonucleotides	ribonucleotides
Final product	DNA	mRNA, rRNA, tRNA
Enzymes that catalyse the formation of the bonds between monomers	DNA Polymerase catalyses the formation of phosphodiester bonds between deoxyribonucleotides	RNA Polymerase catalyses the formation of phosphodiester bonds between ribonucleotides

(c) Explain how complementary base pairing facilitates the storage and transmission of genetic information. [3]

- [Compulsory for Storage] Complementary base pairing between bases of the two template strands / parental strands is important as it **stabilises the double-stranded helix structure** of DNA for storage of genetic information.
- [Transmission] Complementary base pairing enables . . .
- [Replication] the **formation of daughter strands during DNA replication** before mitosis in order to transmit genetic information to daughter cells.
- [Replication] the **proofreading function of DNA polymerase to find and repair mutations** to maintain genetic fidelity.
- [Transcription] the **formation of mRNA during transcription** in order to **transmit information** for the synthesis of **the primary sequence of polypeptides** in protein synthesis. [Mention of the type of genetic information – mark once in either this point or the next]
- [Translation] Complementary base pairing **between mRNA codons and tRNA anticodons during translation** transmits information for the primary sequence of polypeptides in protein synthesis.

[Total: 9]

QUESTION 3

The measles virus (MV) is a spherical, non-segmented, single-stranded negative sense RNA virus. The structure of MV is shown in Fig. 3.1.

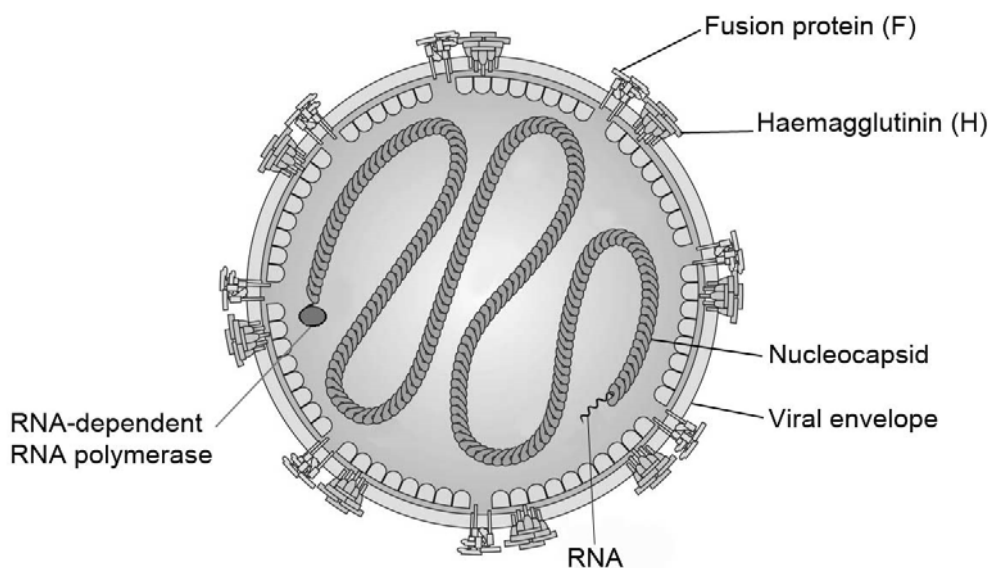


Fig. 3.1

(a) With reference to Fig. 3.1, describe two structural differences between MV and HIV. [2]

- The glycoproteins embedded in the envelope of MV consist of Haemagglutinin/H and fusion protein/F whereas in HIV they are gp120 and gp41.
- HIV contains a capsid surrounding the nucleocapsid whereas MV does not.
- HIV carries two copies of linear (+) single-stranded RNA for its genome whereas MV carries one copy of (-) single-stranded RNA.
- HIV contains reverse transcriptase whereas MV contains RNA-dependent RNA polymerase.

MV only infects cells that have a membrane glycoprotein known as signaling lymphocyte activation molecule (SLAM). When MV infects a cell, **H** acts before **F**. After the virus binds to the host cell, only the nucleoprotein with the viral polymerase enters the host cell and the virus is replicated.

(b) With reference to **Fig. 3.1** and the information provided, suggest how MV infects a cell with SLAM glycoproteins. [3]

- **Haemagglutinin/H** is complementary in shape to the binding site of SLAM receptor
- **Fusion protein /F** causes fusion of the viral envelope to the cell surface membrane
- which releases nucleoprotein and viral polymerase for viral replication

Both MV and HIV infect cells of the immune system. Upon infection, MV causes highly contagious measles which is an airborne disease spreads through the coughs and sneezes of those infected.

(c) (i) Explain how HIV infection causes diseases. [4]

- **Idea that** HIV infects and kills T helper cells [all descriptions about the death of T-cells fall under this point].
- **Idea that** **B lymphocytes cannot produce antibodies** without the help of T lymphocytes.
- This **compromise the immune system**. Leading to opportunistic infections
- **Integration of viral DNA** into proto-oncogene of the host genome, which may also cause cancer.

(ii) Suggest why MV is transmitted at a faster rate as compared to HIV. [1]

- The mode of transmission of MV is **through aerosol/droplet** (OWTTE) hence spread more easily whereas HIV is transmitted through **transfer of body fluids** (OWTTE).

[Total: 10]

QUESTION 4

(a) Telomerase is a ribonucleoprotein which comprises telomerase reverse transcriptase (TERT) protein and telomerase RNA.

Outline how telomerase is formed. [4]

- **Telomerase RNA** is produced through transcription of the telomerase RNA genes in the nucleus (must differentiate between Telomerase RNA gene and TERT gene!)
- **Idea that** the telomerase RNA folds into a 3D structure and remain in the nucleus
- Genes of TERT are transcribed in nucleus to form TERT mRNA...
- ...which is translated by free ribosomes in cytosol to form TERT proteins
- TERT protein is transported from the cytoplasm into nucleus via nuclear pore
- Assembly of telomerase RNA and TERT proteins into telomerase in the nucleus.

During human embryonic development, telomerase activity is activated in embryonic stem cells to enable high proliferation rate of the cell. However, the telomerase activity is usually diminished after birth and the level of telomerase activity is absent in most of the somatic cells.

Fig. 4.1 shows the *TERT* promoter in the two types of cells.

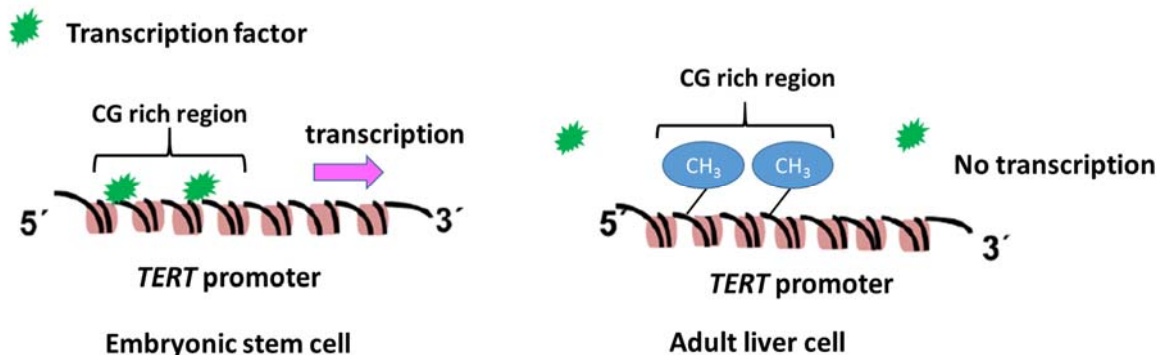


Fig. 4.1

(b) With reference to **Fig. 4.1** and your knowledge, explain why telomerase activity is absent in adult liver cells. [3]

- DNA methylation occur at the CG rich region of the *TERT* promoter
- which recruits histone deacetylase to promote chromatin compaction
- Transcription factors and RNA polymerase cannot access the promoter of *TERT*, hence prevent gene expression

(c) The process occurring in adult liver cells shown in **Fig. 4.1** also occurs in prokaryotic cells.

State how the outcome of the process in prokaryotes differs from that in adult liver cells. [1]

- DNA methylation in adult liver cell leads to long-term inactivation of genes/turning genes off whereas it protects the bacteria DNA from being degraded by restriction enzymes.

(d) Outline the roles of telomeres in eukaryotic cells. [3]

- Protect the organism's genes from being lost with each round of DNA replication.
- Protect the chromosome by binding proteins that prevents the ends from joining to other chromosomes.
- Telomeres and associated proteins prevent the exposed staggered ends of DNA from activating the cell's monitoring system to cause apoptosis of cell.
- Allow the completion of DNA synthesis at the ends of eukaryotic chromosomes.

[Total: 11]

QUESTION 5

To study the inheritance of coat colour and eye colour in deer-mice, scientists performed two crosses and the table below shows the phenotypes of the F₁ generations from these two crosses.

Cross	Parents (pure bred)	F ₁ phenotype	Number of F ₁ progeny
1	Black eye, coloured female X Pink eye, albino male	All black eye, coloured mice	77
2	Black eye, coloured male X Pink eye, albino female	All black eye, coloured mice	68

The F₁ generation were then interbred and the following F₂ offspring were produced:

Black eye, coloured	295
Black eye, albino	42
Pink eye, coloured	46
Pink eye, albino	33

(a) Explain the purpose of carrying out crosses 1 and 2. [2]

- A reciprocal cross to determine whether the two gene loci for coat colour and eye colour are sex-linked.
- [Context required] The same F₁ results (all black eye, coloured mice) are observed suggesting that the two genes are autosomal /not sex-linked.

(b) Using suitable symbols, draw a genetic diagram to explain the results of F₁ cross. [5]

B represents the dominant allele for black eye
b represents the recessive allele for pink eye
A represents the dominant allele for coloured coat
a represents the recessive allele for albino

[1]

F₁ genotype (2n):

$$\begin{array}{c} B \quad A \\ \hline b \quad a \end{array}$$

[1]

F₁ phenotype:

Black eye, coloured

Selfing F₁:

Black eye, coloured

x

Black eye, coloured

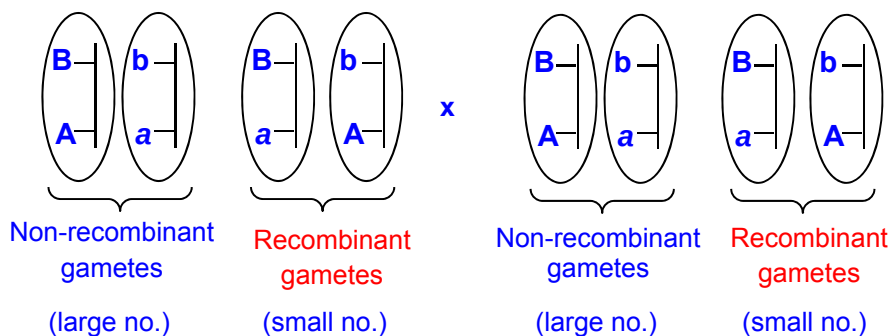
F₁ genotype:

$$\begin{array}{c} B \quad b \\ | \quad | \\ A \quad a \end{array}$$

x

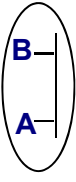
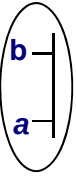
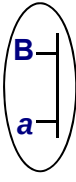
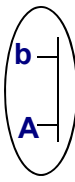
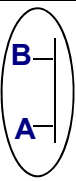
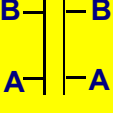
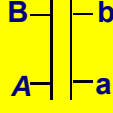
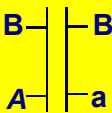
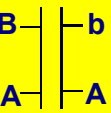
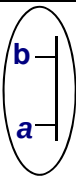
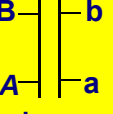
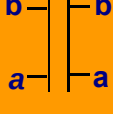
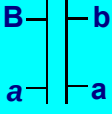
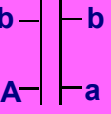
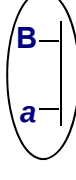
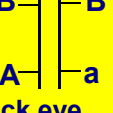
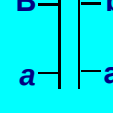
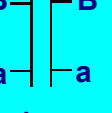
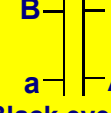
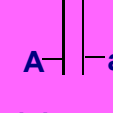
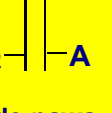
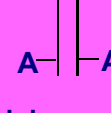
$$\begin{array}{c} B \quad b \\ | \quad | \\ A \quad a \end{array}$$

Gametes:



[1]

genotypes and phenotypes:

	 (Large no.)	 (Large no.)	 (small no.)	 (small no.)
 (Large no.)	 Black eye, coloured	 Black eye, coloured	 Black eye, coloured	 Black eye, coloured
 (Large no.)	 Black eye, coloured	 Pink eye, albino	 Black eye, albino	 pink eye, coloured
 (small no.)	 Black eye, coloured	 Black eye, albino	 Black eye, albino	 Black eye, coloured
 (small no.)	 Black eye, coloured	 pink eye, coloured	 Wide paws, hair	 pink eye, coloured

[1]

Match phenotype with genotype

F₂
phenotypes:

Black eye,
coloured pink eye,
albino

Non-recombinant phenotypes
(large numbers)

Black eye,
albino pink eye,
coloured

Recombinant phenotypes
(small numbers)

[1]

Observed no.

295 : 33

42 : 46

(c) State the expected ratio of the F₂ phenotypes if Mendelian law applies to the two gene loci. [1]

9:3:3:1

The chi-squared (χ^2) test was performed on these results, giving a calculated value for χ^2 of 47.527.

The χ^2 distribution table and equation to calculate χ^2 is shown below.

number of degrees of freedom (v)	probability
	0.05
1	3.84
2	5.99
3	7.82
4	9.49

(d) Use the calculated value of χ^2 and the table of probabilities provided in the table above, explain the conclusions drawn from the (χ^2) test. [3]

- The calculated χ^2 value of 47.527 is greater than the **critical value of 7.82 at 5% significance level**
- The **probability** that the difference between expected and observed number is due to chance is less than 0.05.
- Difference is significant / not due to chance, the observed number of 295:42:46:33 does not conform to the expected ratio 9:3:3:1, hence the two gene loci are linked.

[Total: 11]

QUESTION 6

Two groups of white mustard plants, *Sinapis alba*, were grown, one group under high illumination, the other under low illumination. When fully grown, the effect of increasing light intensity on the rate of photosynthesis in the two groups of plants was measured.

Fig. 6.1 shows the results.

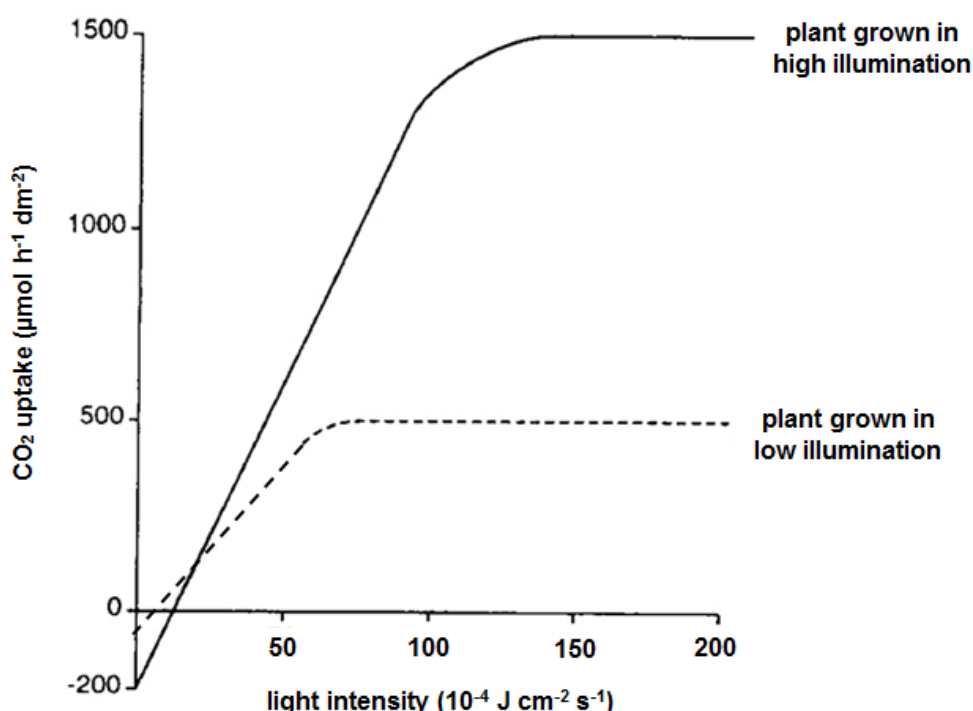


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) state and explain the effect of light intensities above $150 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ on the rate of photosynthesis in plants grown in high illumination. [2]

- For light intensity above $150 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$, CO₂ uptake remains constant at $1500 \mu\text{mol h}^{-1} \text{dm}^{-2}$.
- Light saturation is reached /light intensity is no longer the limiting factor.

(ii) state with evidence, **two** ways in which the carbon dioxide uptake of both plants differ at light intensities below $50 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$. [4]

Any two:

- For light intensity between $20-50 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$, (rate of) CO₂ uptake for the plants grown in high illumination is higher than that grown in low illumination. [1]
 - Figures – [1]
- The compensation point for plants grown in high illumination occurs at a higher light intensity than those grown in low illumination or vice versa [1]
 - High illumination: $15 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ and Low illumination: $7 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ – [1]

- Below the compensation point, plants grown at high illumination give out more carbon dioxide than plants grown in low illumination. [1]
 - High illumination: $-200 \mu\text{mol h}^{-1}\text{dm}^{-2}$ and Low illumination: $-50 \mu\text{mol h}^{-1}\text{dm}^{-2}$ [1]
- Below the compensation point, plants grown at high illumination give out more carbon dioxide than plants grown in low illumination. [1]
 - High illumination: $-200 \mu\text{mol h}^{-1}\text{dm}^{-2}$ and Low illumination: $-50 \mu\text{mol h}^{-1}\text{dm}^{-2}$ [1]
- *(not a good answer)* rate of CO_2 uptake increases at a faster rate for plant grown in high illumination than plants grown in low illumination. [1]
 - High illumination: $-200 \mu\text{mol h}^{-1}\text{dm}^{-2}$ to $650 \mu\text{mol h}^{-1}\text{dm}^{-2}$ and Low illumination: $-50 \mu\text{mol h}^{-1}\text{dm}^{-2}$ to $400 \mu\text{mol h}^{-1}\text{dm}^{-2}$ [1]

(b) Outline the fate of each product of photolysis in the light dependent reaction. [3]

1. Oxygen used by cells for aerobic respiration
2. Excess oxygen is released out of plant through stomata
3. Protons diffuse through ATP synthase **from thylakoid space to stroma** to generate ATP
4. protons combine with electrons from PS-I and **NADP to form NADPH**
5. Electrons are used to **replace electrons loss from PS-II**
6. Electrons are transported along **electron transport chain** by **electron carriers** of progressively lower energy levels
7. In cyclic photophosphorylation, **electron from PS-I goes back to PS-I**
8. In non-cyclic photophosphorylation, electrons from PS-I **combine with protons and NADP to form NADPH**

[Total: 9]

QUESTION 7

Fig. 7.1 is an electron micrograph of a section through a myelinated neurone showing the Schwann cell and axon membrane.

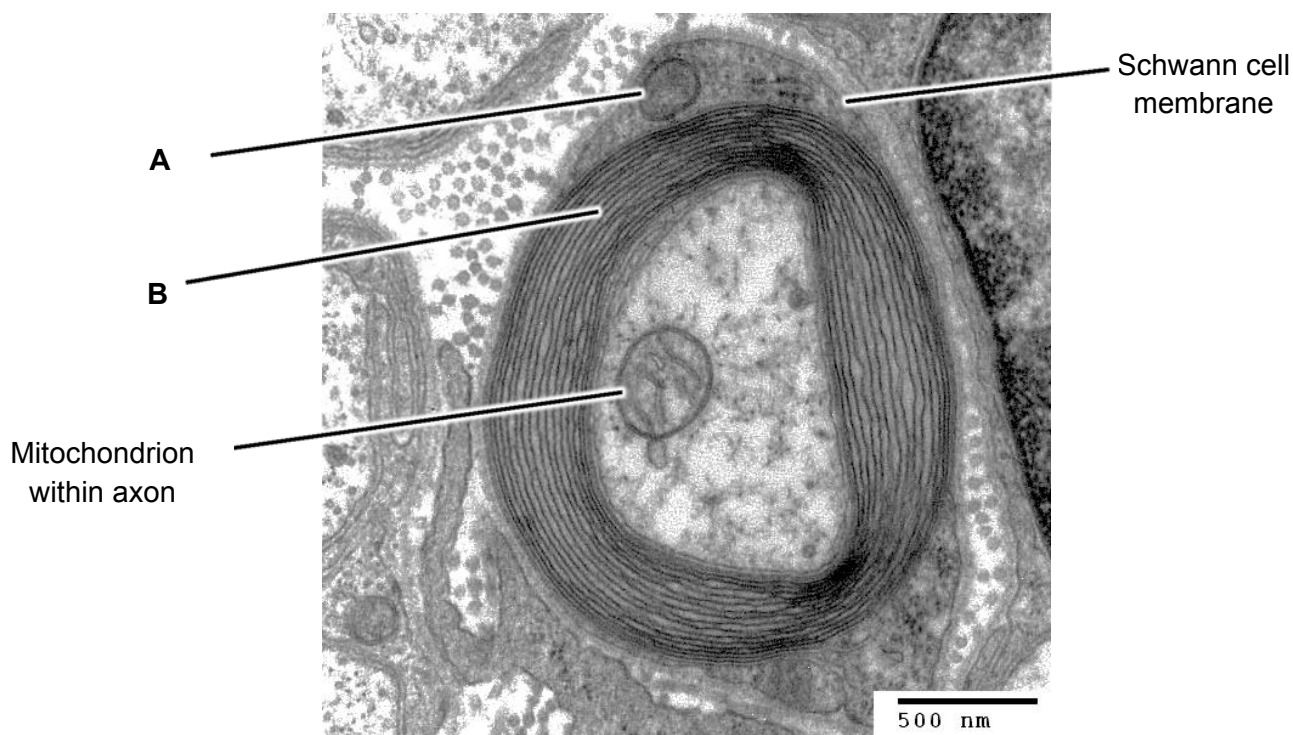


Fig. 7.1

(a) Identify the structures labelled **A** and **B**.

[2]

A Nucleus of the Schwann cell

B Myelin sheath

(b) **Fig. 7.2** shows the percentage of energy used for various processes involved in the maintenance of resting potentials and in the reception and transmission of action potentials by a neuron.

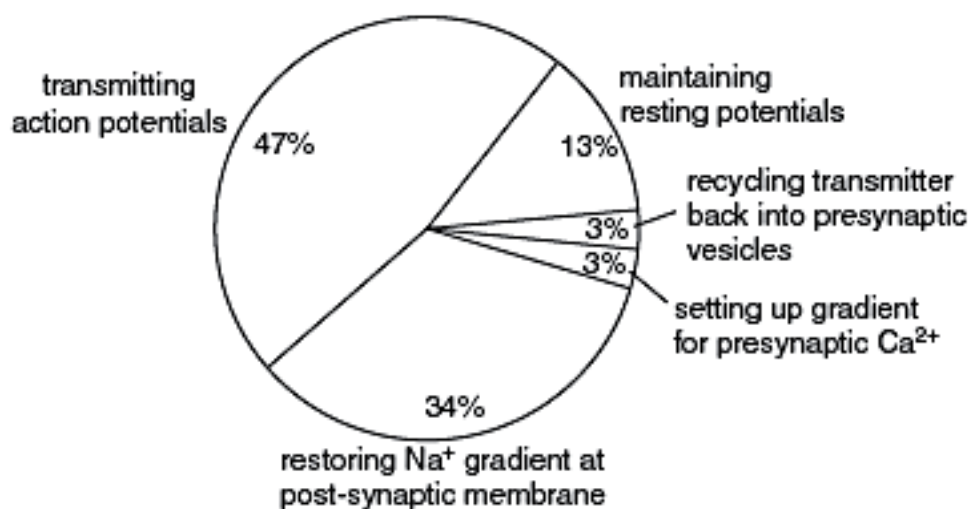


Fig. 7.2

(i) Explain why maintaining a resting potential requires energy. [2]

- Active transport of **3 sodium ions out and 2 potassium ions in** via the **sodium potassium pump**
- **..against concentration gradient** which requires hydrolysis of **ATP**

(ii) Neurons contain large numbers of mitochondria. There are more mitochondria in each dendrite than in the axon.

With reference to **Fig. 7.2**, suggest reasons for the distribution of mitochondria. [3]

- Mitochondria is the site for **ATP synthesis**
Restoring Na^+ gradient at post-synaptic membrane uses **34% energy** which occurs in **dendrites**
- Recycling neurotransmitter **and** setting up Ca^{2+} gradient requires only **6% energy in terminal end of axons**

(c) Describe the role of Ca^{2+} in the passage of impulses across a synapse. [2]

- **Ca^{2+} influx into presynaptic neuron** via facilitated diffusion through voltage-gated Ca^{2+} channels
- The increase in Ca^{2+} causes **vesicles that contain neurotransmitters** move to and **fuse** with **presynaptic membrane**
- causes the **neurotransmitter to be released into the synaptic cleft** by **exocytosis**

(d) Synapses slow down the rate of transmission of nerve impulses but serve other important roles in the nervous system.

Outline two roles of synapses in the nervous system. [2]

Ensure **one-way transmission**

- **Filter out infrequent impulses/** temporal summation
- Allow spatial summation/ convergence of impulse/ interconnection of many nerve cells
- Allow transmission of information between neuron
- Prevent overstimulation

[Total: 11]

QUESTION 8

Mole rats, *Spalax ehrenbergi*, are mammals that live in groups in underground burrows. They are blind, and communicate with each other through sound and scent. Males make a purring call when they attempt to persuade females to mate with them.

In Israel, the mole rats found in different parts of the country all look identical. However, there are actually four different populations with different chromosome numbers, which live in different climatic regions.

Table 8.1 shows the four populations of mole rats and information about the purring calls used by the males in each population. The call of the males were analysed by measuring the number of sound pulses per second, and also the frequencies of the sounds that they made.

Chromosome number of population		52	54	58	60
Climatic region in which population lives		Cool and humid	Cool and dry	Warm and humid	Warm and dry
Purring call made by males	Mean number of pulses per second	21.0	25.3	23.9	23.2
	Mean major frequency/ kHz	595	555	583	562

Table 8.1

Researchers investigated how female mole rats from each of the four populations responded to purring calls made by males from the same population, and by males from different populations.

A female was placed midway between two loudspeakers, and recorded calls from two males were played to her simultaneously. The researchers noted which loudspeaker the female moved towards. This was repeated with many different females from each population.

The results are shown in **Table 8.2**.

Population chromosome number	Percentage of females preferring the purring call of males from their own population
52	79
54	77
58	78
60	44

Table 8.2

(a) With reference to **Table 8.2**, describe the extent to which female mole rats show a preference for the purring calls of males from their own population. [2]

- **High percentage of 79%, 77% and 78% females from populations with 52, 54 and 58 chromosome number respectively prefer calls from their own population**
- **Low percentage of 44% of females from population with 60 chromosome number prefer calls from their own population**

(b) With reference to the data in both **Table 8.1** and **Table 8.2**, discuss whether these four populations of mole rats should be classified as different species. [4]

- **Yes**
- mole rats with different chromosome numbers
- **cannot interbreed to form fertile and viable offspring**
- as not all chromosomes will be able to pair up in meiosis/prophase 1/ idea of producing offspring with odd no. of chromosome (OWTTE)
- Geographically isolated/ live in different habitats so unlikely to interbreed
- Most females prefer males from their own population due to differences in mating call

The phylogenetic relationship of seven genera of mole rats was investigated using nucleotide sequences of the *12S rRNA* gene obtained from mitochondrial DNA.

Fig. 8.1 shows a phylogenetic tree of the mole rats based on this *rRNA* gene nucleotide sequence data.

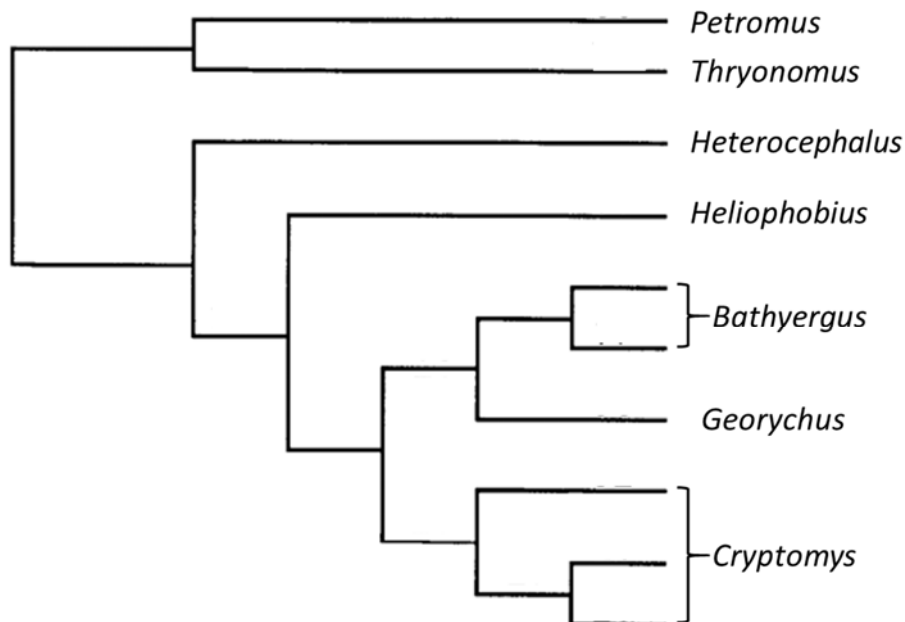


Fig. 8.1

(c) Describe the advantages of using nucleotide data such as the *12S rRNA* gene in classifying the mole rats. [3]

- The mole rats share the **same kind of genetic material** — **DNA**. Hence it is a good basis of comparison.
- It is **objective** as **homologous nucleotide sequence** can be **compared**. It is also **quantitative** as the **differences** can be **counted** and subjected to **statistical analysis**.
- Nucleotide sequence comparison is more complete as it takes into consideration of **silent mutations and changes in non-coding sequences** (not expressed in phenotype)...
- The **rate of accumulation of mutation** in the gene coding for 12s RNA occurred at a **constant rate** through time
- The number of nucleotide differences can be used as a gauge to estimate when **two species diverged from a common ancestor**
- *12S rRNA* gene is found on mtDNA which **lacks germline recombination** (crossing over, independent assortment). Thus, variation in the cytochrome b gene sequence is largely by mutation.
- The probability of recovery of mtDNA from very small or degraded biological samples is higher than the one of nuclear DNA, because the mitochondrial DNA molecules exist in **thousands of copies per cell**, while nuclear DNA has only two copies per cell.
- mtDNA has a **high level of variability** (due to high rate of mutation) in the **non-coding sequence** (control region) which can be used to elucidate phylogenetic relationships among **recently diverged species**. This high rate of mutation of mtDNA is due to the absence of DNA repair mechanism in mitochondria.

[Total: 9]

QUESTION 9

(a) Describe the role of vesicles in a cell.

[6]

1. For transport of proteins from **Endoplasmic Reticulum to Golgi Apparatus**
2. For transport of proteins from **Golgi Apparatus to Cell Surface Membrane**
3. For transport of proteins from **Golgi Apparatus** to other cellular organelles or destinations
4. For **fusion** of vesicle with **cell surface membrane** which results in **replenishment of cell surface membrane**
5. For **fusion** of transport vesicle with GA which results in **replenishment of Golgi Apparatus membrane**
6. For **enclosing** foreign particles by the process of **endocytosis**
7. Functions as **lysosomes** that **store hydrolytic enzymes**
[at least 1 function of lysosome]
8. **Cellulose**-containing vesicles fuse to form the **cell plate** during cytokinesis in plant cells
9. Named examples, e.g. insulin (secreted), G-protein coupled receptor (embedded in CSM), proton pumps (embedded in lysosomal membrane), neurotransmitters (exocytosis).

(b) Describe one causative factor of cancer and explain how this factor increases the chances of cancerous growth. [6]

1. **ONE Causative Factor:** Excessive **UV radiation / chemical carcinogens / viruses / inherited genetic factors**
2. Describe: viruses may **insert** their **genetic material / DNA** into the **host cell genome** may disrupt proto-oncogenes or tumour-suppressor genes.

Describe: Excessive UV radiation / ionising radiation may cause **mutations** to proto-oncogenes or tumour-suppressor genes.

Describe: **Exposure to carcinogens** (chemicals that cause cancer) e.g. ethidium bromide, benzo(a)pyrene in cigarette smoke, sodium nitrite in preserved foods may cause mutations

Describe: An individual inheriting an **oncogene** (a mutated proto-oncogene) or a **mutant** allele of a **tumor-suppressor gene** will be one step closer to accumulating the necessary mutations for cancer development.

3. **Mutated proto-oncogenes** code for **hyperactive proteins** that cause uncontrolled cell division.
4. **Mutated tumour suppressor** genes code for **non-functional proteins** that cause uncontrolled cell division.
5. Cancer is a **multi-step process**, which requires **accumulation of mutations in multiple genes**.
6. Mutations in proto-oncogenes and tumour-suppressor genes can cause cell to **bypass cell cycle checkpoints/ dysregulation** of checkpoints.
7. **Damaged DNA / mutations are not repaired / arrested** at the cell cycle checkpoints.

8. leading to the **accumulation of mutations in a SINGLE cell**, resulting in formation of a cancerous cell that undergoes **uncontrolled cell division**.

(c) Describe the differences between the control of gene expression in prokaryotic and eukaryotic cells. [8]

	Eukaryotic genome	Prokaryotic genome
At chromosomal level		
	1. Histone modifications to regulate how compact the DNA region is	No histone modification as DNA not associated with histones
At transcriptional level		
Control at Promoter	2. One promoter for each gene	One promoter occurs for each operon which consist of several functionally related structural genes
Induction/ repression in response to external stimuli	3. In response to external stimuli, transcription factors may bind to regulatory sequences and activate or silence transcription	In response to external stimuli, regulatory proteins bind to control regions for each operon, inducing or repressing transcription
At post-transcriptional level		
	4. Transcription and translation processes are separated due to presence of nuclear membrane	Transcription and translation occur simultaneously due to the absence of a nuclear membrane
	5. post transcriptional modifications (addition of 5'cap, 3' poly-A tail, splicing) occur	No controls at post-transcriptional level
At translational level		
	6. repressor proteins bind to 5' UTR and blocks translation	No repressor proteins bind to the 5' region of mRNA, translation not blocked
At post-translational level		
	7. degradation of proteins by ubiquitin	Ubiquitin is not involved in degradation of proteins
	8. cleavage may occur for some proteins	No cleavage of proteins

[Total: 20]

QUESTION 10

(a) Describe the structure of collagen and how it is related to its function. [8]

1. **Tropocollagen** is formed when **three** collagen **polypeptide chains** wound around each other to give a **triple helix**.
2. Each collagen polypeptide chain is in the shape of a **loosely wound left-handed helix** that wind around the other two.
3. The three strands are linked together by **hydrogen bonds** formed between peptide N-H group of glycine and peptide C=O group of other amino acids on the other strands.
4. The sequence of amino acids of each strand is usually a repeat of
Glycine – Proline – X, or
Glycine – X – Hydroxyproline where X is any other amino acids except glycine
5. The presence of **glycine** at every third amino acid within each polypeptide chain allows **close packing** of the triple helix to form a **tight coil**.
6. Each complete triple helix of tropocollagen interacts with other tropocollagen molecules running parallel to each other by forming **covalent bonds** between the **lysines** in chains lying next to each other.
7. These cross-links hold many tropocollagen molecules side by side, forming **fibrils**, giving rise to **high tensile strength** (mark once for high tensile strength).
8. In collagen **fibrils**, tropocollagens lie **parallel** with **staggered ends**, which would permit them to **overlap** with the tropocollagens in adjacent fibrils.
9. Aggregation of overlapping collagen **fibrils** form strong collagen **fibers**.
10. **[Function]** Hence collagen is able to **provide structural support** for skin, tendons, cartilage, bones, teeth and connective tissue of blood vessels.

(b) Explain why antibiotic resistance spreads so rapidly among bacteria. [6]

High rate of DNA replication

- Idea that bacteria reproduces rapidly/ frequent DNA replication
- Higher chances for mutation

Mutation transferred to another bacteria by genetic recombination

- Mutation may be on plasmid which can be transferred to the recipient bacterial cell via conjugation ; brief description of **conjugation** [max 2]
- Describe process of **transformation** [max 2]
- Describe process of **transduction** [max 2]

Natural selection

- Those bacteria with antibiotic resistance gene are at **selective advantage** when subjected to antibiotic (selection pressure)
- ..able to survive to reproduce and passed on the mutation/ antibiotic resistance gene to large no. of offspring

(c) Describe the main differences between glucagon and insulin signalling in liver cells. [6]

Stages	Insulin	Glucagon
Reception	1. Tyrosine kinase receptor;	1. G-protein coupled receptor
Transduction	2. Dimerization 3. Phosphorylation of tyrosine residues 4. Activation of relay proteins	2. Receptor activated and changes shape 3. Activates G protein 4. Activation of adenyl cyclase to form cAMP/activates kinases
Response	5. Facilitates transport of glucose into cells 6. Increasing number of glucose carriers 7. Synthesis of glycogen from glucose	5. Facilitates release of glucose out of cells 6. No Increase in number of glucose carriers 7. Breakdown of glycogen by glycogen phosphorylase

[Total: 20]

• END OF PAPER 2 •